SCIENCE & NATURE

Paper Bridge Lab

Ages 9–10 · 30 min · Adult nearby

Mission: Make predictions, run fair tests, and record results to discover how shape changes strength.

Pack your kit

- ☐ 4 sheets of printer paper
- ☐ Two stacks of books the same height
- ☐ About 30 coins or small washers
- ☐ Ruler
- ☐ Pencil and paper for a results table

⚠ Stay safe

- Coins and washers are a choking hazard — keep them away from toddlers and pets.
- Build on a table or the floor, not on a high or wobbly surface.
- No scissors are needed. If you use them, use child-safe scissors.

Trail steps

- ☐ **1.** Make two book towers about 15 cm (6 in) apart.
- ☐ **2.** Lay a flat sheet across the gap. Predict how many coins it will hold, then add coins one at a time to the middle until it sags to the table.
- ☐ **3.** Fold a new sheet into a zigzag (accordion) or a U-shaped channel. Predict, then test the same way.
- ☐ **4.** Invent one design of your own. Record every test in a table: design, prediction, coins held.
- ☐ **5.** Circle your strongest design and write one sentence about why you think it worked.

Think about it

How did folding the paper change its strength? What would you test next?

Skills: Making predictions · Running a fair test · Recording results

Grown-up check: _____